

بسم الله الرحمن الرحيم

King Abdulaziz University
Faculty of Computing and information Technology
Computer Science Department



CPCS-214 Computer Architecture and Organization

41 - Pipelined Control

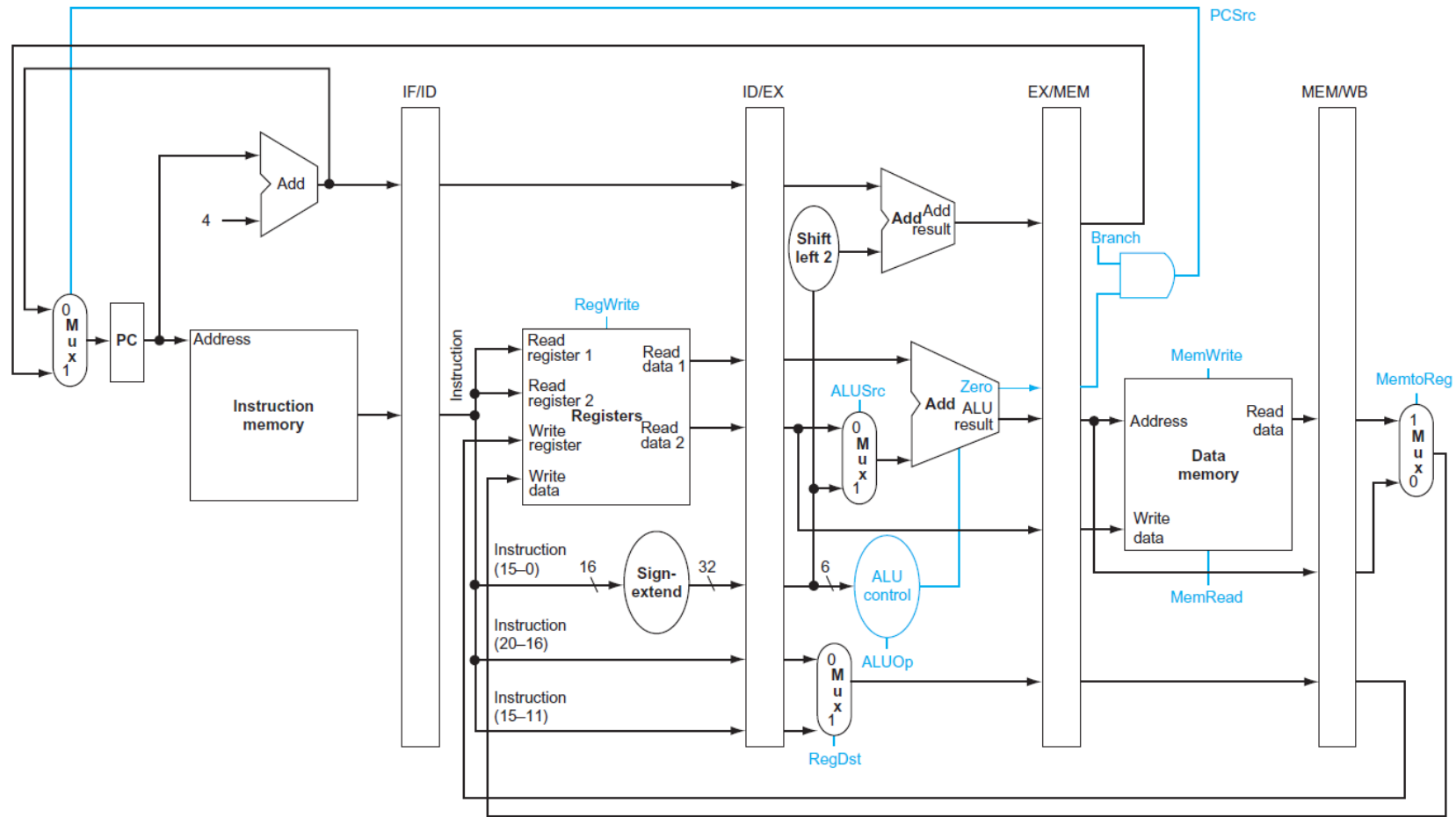
Introduction

- Add control to the pipelined datapath

Control Lines

- Label control lines on the existing datapath
- PC is written on each clock cycle,
 - So there is no separate write signal for PC
- No separate write signals for the pipeline registers (IF/ID, ID/EX, EX/MEM, and MEM/WB)
 - Since pipeline registers are also written during each clock cycle

Control Lines



Stages Control Lines

- To specify control for pipeline, we need only set control values during each pipeline stage
- Because each control line is associated with a component active in only a single pipeline stage, we can divide the control lines into five groups according to pipeline stage

Instruction Fetch

- Control signals to read instruction memory and to write the PC are always asserted
- So there is nothing special to control in this pipeline stage

Instruction Decode/Register File Read

- As in the previous stage, the same thing happens at every clock cycle
- So there are no optional control lines to set

Execution/Address Calculation

- Signals to be set are
 - RegDst
 - ALUOp
 - ALUSrc
- Signals select
 - Result register
 - ALU operation
 - Either Read data 2 or a sign-extended immediate for ALU

ALU Control Bits

Instruction opcode	ALUOp	Instruction operation	Function code	Desired ALU action	ALU control input
LW	00	load word	XXXXXX	add	0010
SW	00	store word	XXXXXX	add	0010
Branch equal	01	branch equal	XXXXXX	subtract	0110
R-type	10	add	100000	add	0010
R-type	10	subtract	100010	subtract	0110
R-type	10	AND	100100	AND	0000
R-type	10	OR	100101	OR	0001
R-type	10	set on less than	101010	set on less than	0111

Memory access

- Control lines set in this stage are
 - Branch
 - MemRead
 - MemWrite
- Beq, load, and store instructions set these signals
- PCSrc selects next sequential address unless control asserts Branch and the ALU result was 0

Control Signals

Signal name	Effect when deasserted (0)	Effect when asserted (1)
RegDst	The register destination number for the Write register comes from the rt field (bits 20:16).	The register destination number for the Write register comes from the rd field (bits 15:11).
RegWrite	None.	The register on the Write register input is written with the value on the Write data input.
ALUSrc	The second ALU operand comes from the second register file output (Read data 2).	The second ALU operand is the sign-extended, lower 16 bits of the instruction.
PCSrc	The PC is replaced by the output of the adder that computes the value of PC + 4.	The PC is replaced by the output of the adder that computes the branch target.
MemRead	None.	Data memory contents designated by the address input are put on the Read data output.
MemWrite	None.	Data memory contents designated by the address input are replaced by the value on the Write data input.
MemtoReg	The value fed to the register Write data input comes from the ALU.	The value fed to the register Write data input comes from the data memory.

Write-Back

- Two control lines are
 - **MemtoReg**, which decides between sending ALU result or memory value to register file
 - **Reg-Write**, which writes the chosen value

Control Lines Grouping

- Since pipelining datapath leaves meaning of the control lines unchanged, we can use the same control values
- Nine control lines are grouped by pipeline stage

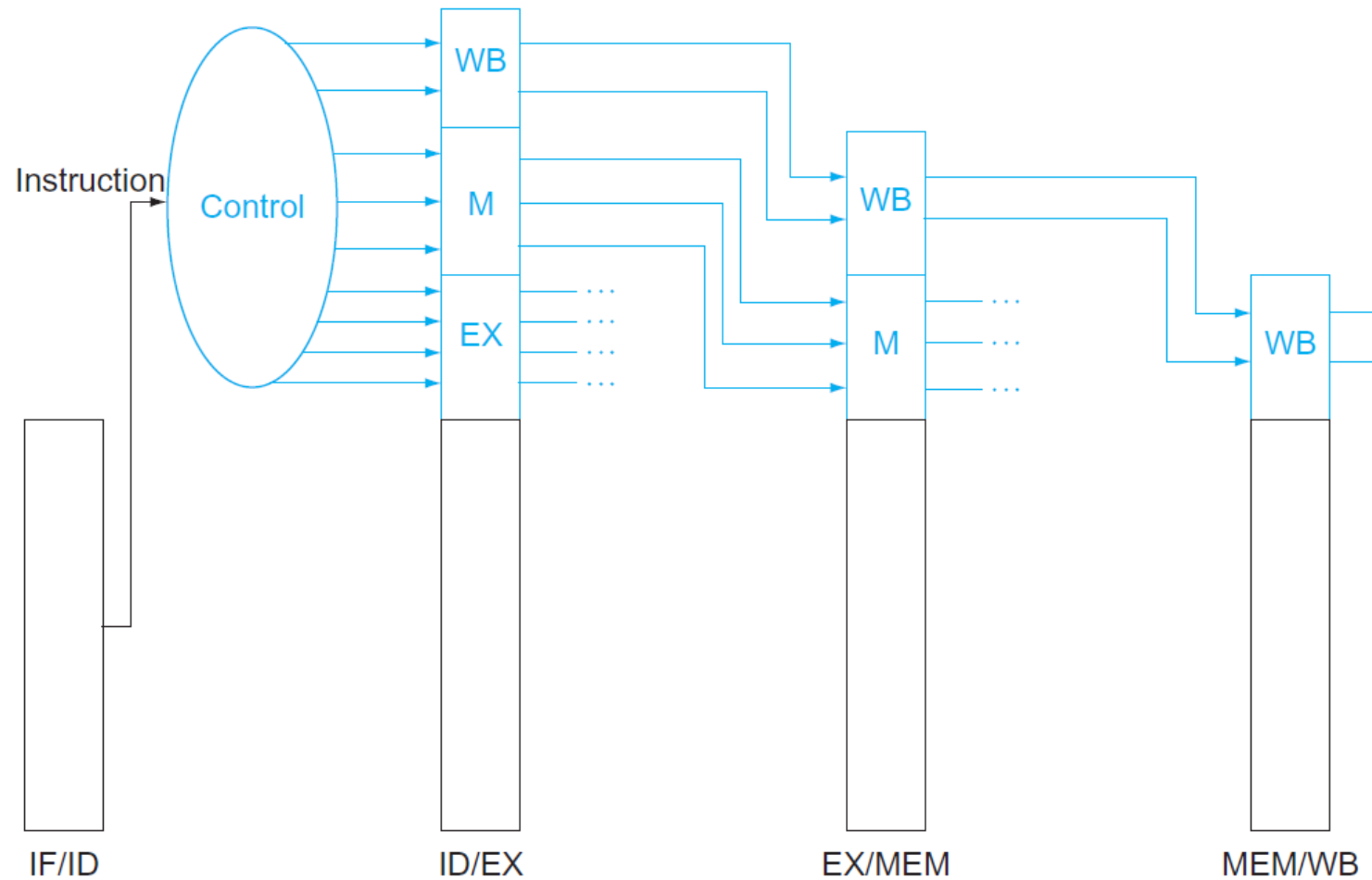
Control Lines Grouping

Instruction	Execution/address calculation stage control lines				Memory access stage control lines			Write-back stage control lines	
	RegDst	ALUOp1	ALUOp0	ALUSrc	Branch	Mem-Read	Mem-Write	Reg-Write	Memto-Reg
R-format	1	1	0	0	0	0	0	1	0
lw	0	0	0	1	0	1	0	1	1
sw	X	0	0	1	0	0	1	0	X
beq	X	0	1	0	1	0	0	0	X

Control Implementation

- Implementing control means setting the nine control lines to these values in each stage for each instruction
- Simplest way to do this is to **extend pipeline registers** to include control information
- Since control lines start with EX stage, we can create control information during instruction decode

Control Implementation



Pipeline Datapath and Control

- Figure shows full datapath with extended pipeline registers and with control lines connected to the proper stage

Pipeline Datapath and Control

